

Chapter 3. Research Protocol

3.1 Title

Influence of Deep Tech-Based Digital Transformation on Quality and Efficiency of University Education in Emerging Contexts

This study investigates how Deep Technology-based digital transformation influences educational quality and institutional efficiency in universities operating within emerging economies through an explanatory sequential mixed methods approach.

3.2 Abstract

Deep Technologies—including Artificial Intelligence, Big Data Analytics, Cloud Computing, Blockchain, Internet of Things, and Digital Twins—are transforming higher education by enabling intelligent learning environments and data-driven institutional governance. Despite increasing investments in digital transformation, universities in emerging contexts continue to experience significant challenges related to educational quality and operational efficiency. This research aims to analyze how Deep Tech-based digital transformation influences these two dimensions by integrating quantitative institutional indicators with qualitative stakeholder perspectives. Using an explanatory sequential mixed methods design, the study seeks to develop an evidence-based framework supporting sustainable digital transformation strategies for higher education institutions in emerging economies.

3.3 Introduction and Problem Statement

Higher education institutions worldwide are undergoing profound digital transformation driven by the rapid development of Deep Technologies. Universities increasingly integrate Artificial Intelligence, learning analytics, intelligent automation, cloud-based educational platforms, blockchain credentials, and predictive analytics into teaching, research, and institutional governance. These technologies promise to improve learning outcomes, optimize administrative processes, and strengthen institutional competitiveness.

However, universities operating within emerging economies continue to face significant structural constraints, including limited technological infrastructure, unequal digital access, financial restrictions, and heterogeneous digital

competencies. Consequently, the implementation of Deep Technologies frequently produces inconsistent institutional outcomes. While some universities report measurable improvements in educational quality and operational efficiency, others experience limited benefits despite similar technological investments.

Current literature primarily examines digital transformation, educational quality, or institutional efficiency independently. Few studies explain how these dimensions interact within universities located in emerging contexts or identify the organizational mechanisms through which Deep Technologies generate sustainable educational value. This knowledge gap limits evidence-based decision-making regarding digital transformation strategies in higher education.

Therefore, this study addresses the following research problem:

How does Deep Tech-based digital transformation influence educational quality and institutional efficiency in universities operating within emerging contexts, and which organizational, technological, and human factors explain successful digital transformation?

3.4 Literature Review

The literature review will synthesize current theoretical and empirical evidence regarding the intersection of **Deep Technologies**, **digital transformation**, **educational quality**, and **institutional efficiency** in higher education. Particular attention will be given to studies published between **2020 and 2026**, reflecting the rapid evolution of Artificial Intelligence and emerging digital technologies in university environments.

The review will be organized into five thematic areas. The first examines the conceptual evolution of **Deep Technologies**, including Artificial Intelligence, Machine Learning, Big Data Analytics, Cloud Computing, Internet of Things, Blockchain, Digital Twins, and Intelligent Automation. The second analyzes digital transformation frameworks in higher education, emphasizing organizational change, digital governance, and smart university models. The third explores educational quality from the perspectives of student learning outcomes, teaching innovation, curriculum modernization, and learner engagement. The fourth reviews institutional efficiency, focusing on administrative automation, resource optimization, decision-making processes, and organizational performance.

Finally, the fifth identifies empirical studies conducted in emerging economies, highlighting methodological approaches, reported outcomes, research gaps, and future research directions.

Special emphasis will be placed on identifying the theoretical and empirical gap concerning how Deep Tech-based digital transformation simultaneously influences educational quality and institutional efficiency within universities operating in resource-constrained environments. The review will therefore establish the conceptual foundation for the proposed explanatory framework.

3.5 Research Questions

Following the Mixed Methods paradigm adopted in this study, research questions are formulated instead of statistical hypotheses because the investigation seeks both explanation and contextual understanding.

Main Research Question

How does Deep Tech-based digital transformation influence educational quality and institutional efficiency in universities operating within emerging contexts?

Secondary Research Questions

RQ1. How does the adoption of Deep Technologies affect educational quality in higher education institutions?

RQ2. How does Deep Tech-based digital transformation improve institutional efficiency within universities?

RQ3. Which organizational, technological, and human factors facilitate successful digital transformation in emerging universities?

RQ4. How do university stakeholders perceive the educational and organizational impacts of Deep Technology implementation?

Together, these questions enable the integration of quantitative institutional evidence with qualitative stakeholder perspectives, thereby supporting a comprehensive explanation of digital transformation.

3.6 Methodology

This research adopts an **Explanatory Sequential Mixed Methods Design**, consistent with the methodological decisions established during Sessions 1 and

2. The design begins with quantitative data collection and analysis, followed by qualitative investigation intended to explain the quantitative findings.

Research Paradigm

Mixed Methods Research Paradigm.

Research Design

Explanatory Sequential Mixed Methods Design (QUAN → QUAL → Integration).

Quantitative Phase

The quantitative phase will evaluate the relationship between Deep Technology implementation and institutional performance using measurable indicators of educational quality and operational efficiency. Data will be collected from participating universities through institutional databases, learning management systems, administrative reports, and standardized questionnaires.

Representative indicators include:

- Student academic achievement.
- Student retention and graduation rates.
- Learning analytics indicators.
- Administrative processing time.
- Resource utilization.
- Digital service adoption.
- Institutional productivity.

Descriptive statistics, correlation analysis, multiple regression, and Structural Equation Modeling (SEM), where appropriate, will be employed to identify significant relationships among variables.

Qualitative Phase

The qualitative phase will explain the quantitative findings through semi-structured interviews with university administrators, faculty members, information technology managers, instructional designers, and students.

Interview topics will include:

- Leadership and institutional vision.
- Organizational culture.
- Faculty digital competencies.
- Barriers to technology adoption.
- Student learning experiences.
- Institutional innovation practices.

- Perceived educational benefits.

Interview transcripts will be analyzed using thematic analysis supported by qualitative analysis software such as NVivo or ATLAS.ti.

Integration Phase

The final stage integrates quantitative and qualitative findings through joint interpretation. Quantitative results will identify measurable institutional outcomes, whereas qualitative evidence will explain the organizational mechanisms responsible for those outcomes.

This integration will culminate in the development of an explanatory framework describing how Deep Technologies generate improvements in educational quality and institutional efficiency in emerging universities.

3.7 Ethical Considerations

This research will comply fully with the ethical standards established by the **Universidad Nacional Mayor de San Marcos**, the **National Code of Scientific Integrity of Peru (CONCYTEC Resolution No. 028-2024-CONCYTEC-P)**, and internationally accepted principles of responsible research.

Participation in interviews and surveys will be voluntary. All participants will receive detailed information regarding the objectives of the study and will provide informed consent prior to participation. Confidentiality and anonymity will be guaranteed through data coding procedures, secure digital storage, and restricted access to research data.

Institutional performance data obtained from participating universities will be anonymized before analysis to prevent institutional identification. Results will be reported in aggregated form, ensuring that neither individuals nor universities can be identified without explicit authorization.

The research does not involve vulnerable populations, medical interventions, or experimental manipulation. Consequently, the anticipated level of ethical risk is minimal and primarily concerns confidentiality and responsible data management.

3.8 Expected Results

The study is expected to demonstrate that Deep Tech-based digital transformation positively influences both educational quality and institutional efficiency within universities operating in emerging contexts.

Specifically, the quantitative phase is expected to identify statistically significant relationships between Deep Technology implementation and improvements in student learning outcomes, teaching innovation, administrative productivity, resource optimization, and institutional performance.

The qualitative phase is expected to reveal the organizational conditions that explain successful digital transformation, including effective leadership, institutional digital culture, faculty engagement, technological readiness, and stakeholder collaboration.

The integration of both phases will produce:

- A comprehensive explanatory framework of Deep Tech-enabled digital transformation.
- Evidence-based recommendations for university digital transformation strategies.
- Practical guidelines supporting institutional modernization in emerging economies.
- Scientific publications contributing to the international literature on Artificial Intelligence in higher education.

3.9 Timeline and Budget

The research is planned over a **three-year doctoral period**, following the recommendations presented during Session 3 regarding realistic scheduling, ethics approval, and contingency planning. Approximately **25–30% of the total schedule** has been reserved as a contingency buffer to accommodate ethics review, institutional access, data collection delays, and unforeseen methodological challenges.

Year 1

- Literature review.
- Protocol refinement.
- Ethics approval.
- Instrument development.

- Pilot study.
- Quantitative data collection.

Year 2

- Quantitative data analysis.
- Qualitative interviews.
- Thematic analysis.
- Integration of findings.
- Preliminary conference publications.

Year 3

- Framework validation.
- Scientific article submission.
- Dissertation writing.
- Final revisions.
- Thesis defense.
- Knowledge transfer activities.

The estimated budget will include software licenses, transcription services, travel for institutional visits, conference participation, publication costs, and contingency resources.

3.10 Bibliography

The protocol will be supported by an interdisciplinary body of literature integrating Artificial Intelligence, Deep Technologies, digital transformation, educational quality, organizational management, and mixed methods research.

The bibliography will include:

- Peer-reviewed journal articles published between **2020 and 2026**.
- Methodological references on Mixed Methods Research (Creswell & Creswell).
- Case Study Research (Yin).
- Design Science Research (Hevner et al.).
- Organizational Digital Transformation.
- Artificial Intelligence in Higher Education.
- Smart Universities.
- Educational Analytics.
- Responsible Artificial Intelligence.

- Digital Governance.
- Scientific Integrity regulations issued by CONCYTEC.

The final dissertation will adopt the **APA 7th edition** citation style and will prioritize high-impact international journals indexed in **Scopus** and **Web of Science**, ensuring that both theoretical foundations and methodological decisions are supported by current and authoritative scientific evidence.